

Translated

From: e-maxx.ru

Euclidean algorithm for computing the greatest common divisor

Given two non-negative integers a and b , we have to find their **GCD** (greatest common divisor), i.e. the largest number which is a divisor of both a and b . It's commonly denoted by $\gcd(a, b)$. Mathematically it is defined as:

$$\gcd(a, b) = \max \{k > 0 : (k \mid a) \text{ and } (k \mid b)\}$$

(here the symbol " $\mid$ " denotes divisibility, i.e. " $k \mid a$ " means " k divides a ")

When one of the numbers is zero, while the other is non-zero, their greatest common divisor, by definition, is the second number. When both numbers are zero, their greatest common divisor is undefined (it can be any arbitrarily large number), but it is convenient to define it as zero as well to preserve the associativity of $\gcd$. Which gives us a simple rule: if one of the numbers is zero, the greatest common divisor is the other number.

The Euclidean algorithm, discussed below, allows to find the greatest common divisor of two numbers a and b in $O(\log \min(a, b))$. Since the function is **associative**, to find the GCD of **more than two numbers**, we can do $\gcd(a, b, c) = \gcd(a, \gcd(b, c))$ and so forth.

The algorithm was first described in Euclid's "Elements" (circa 300 BC), but it is possible that the algorithm has even earlier origins.

Algorithm

Originally, the Euclidean algorithm was formulated as follows: subtract the smaller number from the larger one until one of the numbers is zero. Indeed, if g divides a and b , it also divides $a-b$. On the other hand, if g divides $a-b$ and b , then it also divides $a = b + (a-b)$, which means that the sets of the common divisors of $\{a, b\}$ and $\{b, a-b\}$ coincide.

Note that a remains the larger number until b is subtracted from it at least $\left\lfloor \frac{a}{b} \right\rfloor$ times. Therefore, to speed things up, $a-b$ is substituted with $a - \left\lfloor \frac{a}{b} \right\rfloor b = a \bmod b$. Then the algorithm is formulated in an extremely simple way: